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BRAC University (Department of Computer Science and Engineering)
CSE 330 (Numerical Methods)
Summer 2026 Semester
Quiz 6

Student ID:
Section:
Name:

Full Marks: 10 marks

Duration: 15 minutes

You are given the function $f(x) = e^{-x} + \cos(x)$.

- e. Find the numerical integral $I_2(f)$ in the interval $[0.25, 1]$
- f. Derive the first weighted factor, σ_1 for Simpsons rule.

$$\begin{aligned} \text{a) } I_2(f) &= \frac{b-a}{6} \left\{ f(n_0) + 4f(n_1) + f(n_2) \right\} \\ &= \frac{1-0.25}{6} \left\{ f(0.25) + 4f(0.625) + f(1) \right\} \\ &= 0.125 (1.7477 + 4 \times 1.346 + 0.908) \\ &= 1.005 \end{aligned}$$

b) Same as lecture note.